Chemistry

Concise Lecture Notes

Structure of Atom

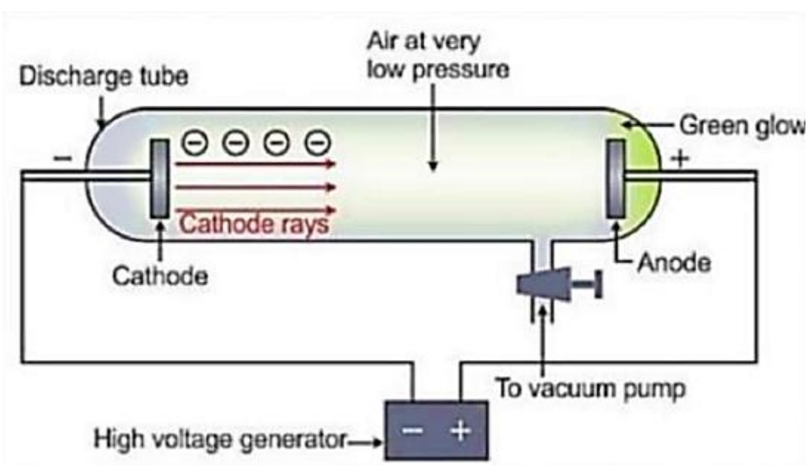
Key topics covered in the Class:

- Discovery of Sub-atomic Particles (Cathode/Anode Rays)
- Atomic Models (Thomson, Rutherford, Bohr)
- Electromagnetic Radiation & Spectrum
- Photoelectric Effect
- Dual Nature of Matter & Uncertainty Principle
- Quantum Numbers & Electronic Configuration

Cathode Ray Experiment

- Cathode rays consist of negatively charged particles called electrons produced in a discharge tube at very low pressure.
- Specific charge: $(e/m) = 1.76 \times 10^{11}\ C\ kg^{-1}$ (remains constant regardless of the gas used).

- Properties include traveling in straight lines, casting shadows, producing mechanical effects, and deflecting towards positive fields.



Millikan's Oil drop Experiment

- Determined the charge of an electron by balancing gravitational force with electric force.
- $q = \frac{mg}{E}$ (where q is charge, m is mass, and E is electric field).
- Established that charge is quantized ($q = ne$), with the elementary charge being -1.6×10^{-19} C.

POSITIVE RAYS- DISCOVERY OF PROTON

- Anode (Canal) rays are positively charged particles produced from the gas between electrodes.

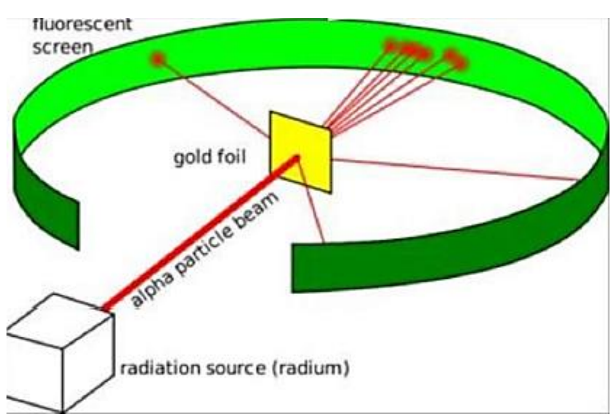
- Unlike cathode rays, their e/m ratio depends on the nature of the gas used (maximum for Hydrogen).
- Discovery of neutrons was later confirmed by Chadwick by bombarding Beryllium with α–particles.

Properties of Charge

- Charge is always quantized ($q = ne$) and exists in two types: positive and negative.
- Coulombic force: $F = K\frac{q_1 q_2}{r^2}$; Potential Energy: $PE = K\frac{q_1 q_2}{r}$.
- Velocity of a charged particle accelerated by voltage V: $v = \sqrt{\frac{2qV}{m}}$.

Rutherford Atomic Model

- The atom consists of a tiny, massive, positively charged nucleus surrounded by electrons in circular orbits.
- Radius of nucleus formula: $R = R_0(A)^{1/3}$ cm (where A is mass number and $R_0 \approx 1.1$ to 1.44×10^{-13} cm).
- **Drawback:** Could not explain stability (Maxwell's theory says accelerated charges radiate energy) or the line spectrum of Hydrogen.



Electromagnetic wave radiation

• Radiation consists of oscillating electric and magnetic fields perpendicular to each other and the direction of propagation.

• **Key Relationship:** $c = \nu\lambda$ (Velocity of light = Frequency × Wavelength).

• Wave number ($\bar{\nu}$) is the reciprocal of wavelength: $\bar{\nu} = \frac{1}{\lambda}$.

Quantum Theory of Light

• Energy is emitted or absorbed in discrete packets called quanta (or photons for light).

• **Energy of a photon:** $E = h\nu = \frac{hc}{\lambda}$ (where Planck's constant $h = 6.626 \times 10^{-34}$ J s).

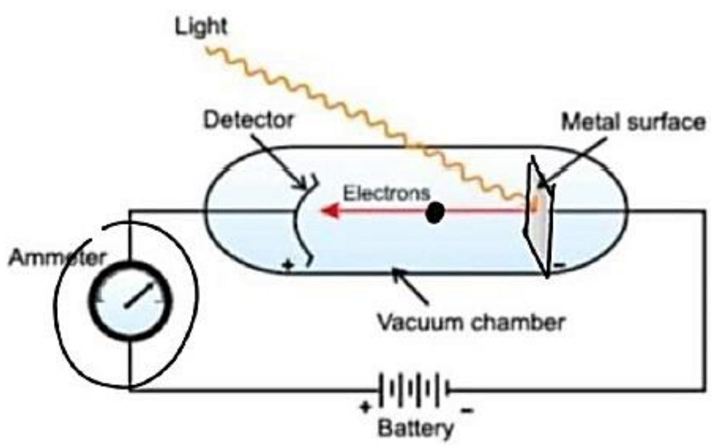
• Explains particle nature phenomena like Black Body Radiation and the Photoelectric Effect.

Photo electric Effect

• Ejection of electrons from a metal surface when light of a frequency higher than the threshold frequency (ν_0) strikes it.

• **Equation:** $h\nu = W + KE_{max}$ (Energy of incident photon = Work Function + Kinetic Energy).

• Number of ejected electrons depends on light intensity; Kinetic energy depends on light frequency.



Bohr's Atomic Model

• Electrons move in stationary circular orbits where angular momentum is quantized: $mvr = \frac{nh}{2\pi}$.

• Energy is absorbed or emitted only when an electron jumps between orbits ($\Delta E = E_{final} - E_{initial}$).

• Applicable primarily to single-electron species (H, He^+, Li^{2+}, etc.).

Radius and Velocity Calculations

- **Radius:** $r_n = 0.529 \frac{n^2}{Z}$ Å (Radius increases with shell number n).
- **Velocity:** $v_n = 2.18 \times 10^6 \frac{Z}{n}$ m s^{-1} (Velocity decreases as shell number n increases).
- Z is the atomic number and n is the orbit number.

Energy Calculations

- Total Energy (TE) of electron in n^{th} orbit: $E_n = -13.6 \frac{Z^2}{n^2}$ eV/atom.
- **Relations:** Total Energy = −Kinetic Energy = $\frac{1}{2}$ Potential Energy.
- **Ground State:** The lowest energy state $(n = 1)$; **Excited State:** Any state where $n > 1$.

Spectrum

- **Emission Spectrum:** Bright lines on a dark background; **Absorption Spectrum:** Dark lines on a bright background.
- **Hydrogen Series:** * Lyman $(n_1 = 1,$ UV region) * Balmer $(n_1 = 2,$ Visible region) * Paschen/Brackett/Pfund $(n_1 = 3,4,5,$ Infrared region).
- **Rydberg Formula:** $\bar{v} = \frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$.

No. of photons emitted by a sample of H atom

- Total unique spectral lines when electron drops from n to 1: $\frac{n(n-1)}{2}$.
- Total unique lines for transition between n_2 and n_1: $\frac{\Delta n(\Delta n+1)}{2}$ (where $\Delta n = n_2 - n_1$).

Dual nature of electron (de-Broglie Hypothesis)

- Matter possesses both wave and particle characteristics.
- **de-Broglie Wavelength:** $\lambda = \frac{h}{mv} = \frac{h}{p} = \frac{h}{\sqrt{2m(KE)}}$.
- For an electron accelerated by potential V: $\lambda = \frac{12.3}{\sqrt{V}}$ Å.

Heisenberg's Uncertainty Principle

- It is impossible to measure the exact position and momentum of a fast-moving particle simultaneously.
- **Formula:** $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$ (where Δx is uncertainty in position and Δp is uncertainty in momentum).

Orbital

- The 3D region around the nucleus where the probability of finding an electron is maximum (90-95%).
- **Shapes:** s-orbital (spherical), p-orbital (dumbbell), d-orbital (double dumbbell).
- **Nodes:** Regions of zero probability. * Radial Nodes = $n - l - 1$ * Angular Nodes = l

Quantum Numbers

- **Principal (n):** Shell size and energy ($n = 1,2,3 \ldots$). Max electrons = $2n^2$.
- **Azimuthal (l):** Subshell shape (0 to $n-1$). $s = 0, p = 1, d = 2, f = 3$. Orbital angular momentum = $\frac{h}{2\pi}\sqrt{l(l+1)}$.
- **Magnetic (m):** Orientation ($-l$ to $+l$). Total orbitals = $2l + 1$.
- **Spin (s):** Spin direction ($\pm 1/2$). Spin magnetic moment = $\sqrt{n(n+2)}$ B.M.

Electronic Configuration Rules

- **Aufbau Principle:** Electrons fill orbitals of lower energy first. Energy order is determined by $(n + l)$ rule.
- **Pauli's Exclusion Principle:** No two electrons in an atom can have the same set of four quantum numbers.
- **Hund's Rule:** Pairing in orbitals doesn't occur until each orbital in the subshell is singly occupied.

Stability of Orbitals

- Half-filled (p^3, d^5) and fully-filled (p^6, d^{10}) configurations are extra stable.
- **Reasons:** Symmetrical distribution of electrons and high Exchange Energy.
- **Exchange Energy:** Energy released when electrons with the same spin exchange positions in degenerate orbitals.

Quantum Mechanical Model

- Based on the Schrödinger wave equation: $\hat{H}\psi = E\psi$.
- ψ (Psi): Wave function (amplitude); ψ^2: Probability density (probability of finding electron).
- Probability of finding electron in a spherical shell: $4\pi r^2 dr \psi^2$.